UNIFIED ASSET & OPERATIONS BRAIN

AI-Powered Industrial Knowledge Intelligence Platform

A Unified MVC System featuring NLP Ingestion, 2D Force-Directed Knowledge Graph, Expert RAG Copilot, Causal Failure Diagnostics (RCA), and Safety Compliance Auditing.

LIVE PROTOTYPE DEPLOYMENT

https://industrial-operation-brain.onrender.com/

GITHUB REPOSITORY SOURCE

https://github.com/ParthSharma532007/Industrial-Operation-Brain

1. Executive Summary

Heavy asset industries (such as oil & gas refining, chemical manufacturing, and electricity generation plants) suffer from severe operational inefficiencies due to information fragmentation. Technical documentation like Standard Operating Procedures (SOPs), Piping & Instrumentation Diagrams (P&IDs), regulatory manuals, and maintenance logs reside in disconnected databases. When senior operators retire, decades of undocumented tribal knowledge vanish.

The Unified Asset & Operations Brain solves this challenge by implementing an intelligent knowledge-engineering platform. It ingests legacy procedures, extracts key equipment tags (e.g. V-202) and operating limits using a rule-based NLP pipeline, links them into an interactive 2D Force-Directed Knowledge Graph, and provides plant engineers with an expert RAG Copilot, FMEA-guided diagnostics (RCA), and automated compliance package builders.

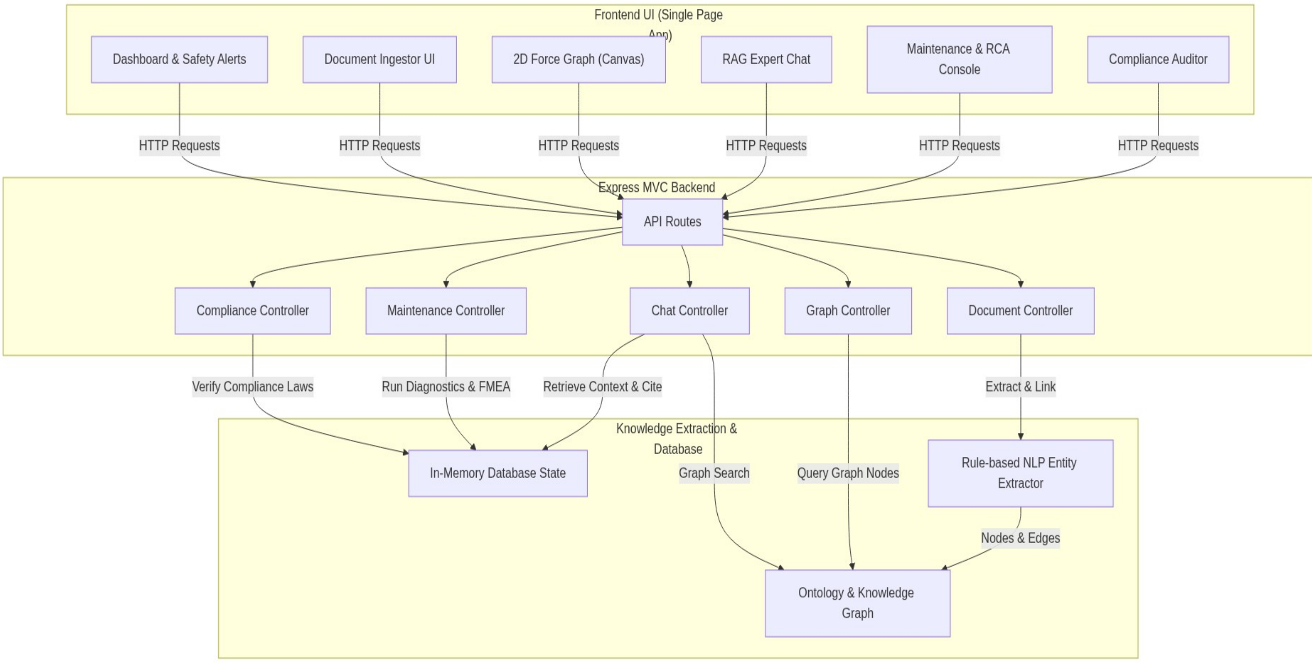
2. The Problem Context

• 35% Working Hours Lost: Engineers waste more than a third of their shift searching for specific historical context or safety guidelines across siloed legacy repositories.

• 18% to 22% Unplanned Outages: Plant downtime is compounded because field technicians lack instant troubleshooting checklists or calibration history at the equipment location.

• Compliance & Safety Risk: Maintaining compliance with factory safety standards and local regulations requires hours of manual testing log audits, exposing plants to significant liability.

3. Platform Architecture Diagram



4. Core System Workflow

1. Technical SOPs/Blueprints are pasted into the Ingestor UI -> Controller routes payload to NLP service.

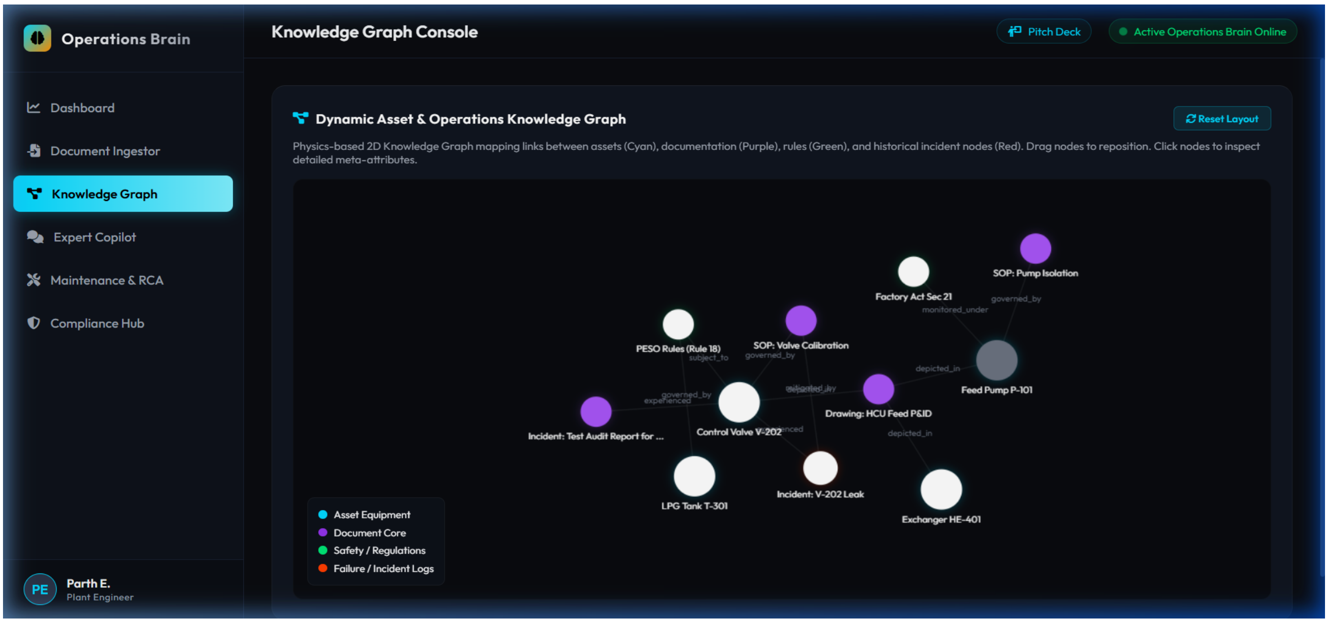
2. The NLP Service identifies asset tags and physical boundary parameters and instantiates graph nodes.

3. When an engineer queries the Expert Copilot, the RAG logic combines dense keyword vectors and graph topology coordinates to formulate the answer and cite exactly where the data is in the files.

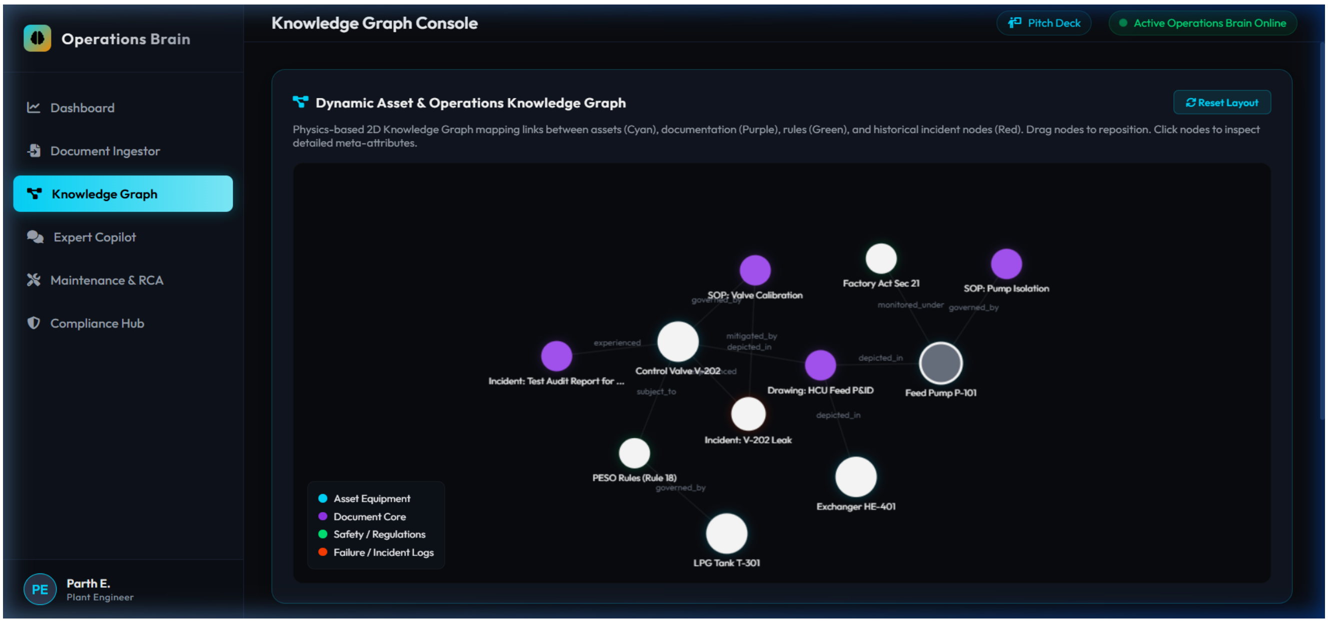
4. Asset sensors are audited against compliance tables (e.g. PESO) and FMEA guidelines to compile RCA logs.

5. Visual Demonstration

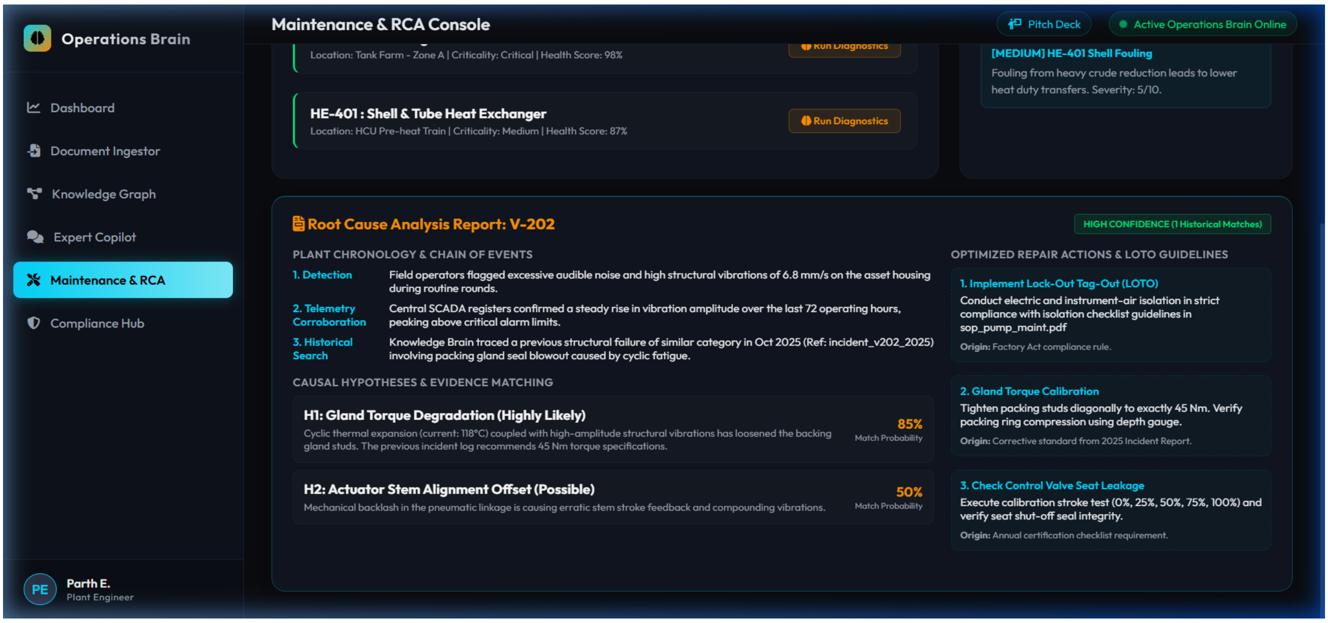
Interactive 2D Force-Directed Knowledge Graph:



Selected Node Detail Inspector Overlay:



6. RCA & Failure Analysis Engine



7. Future Roadmap

• Intelligent P&ID Digitization: Deploy computer vision pipelines to parse scanned piping diagrams and automatically identify instrument junctions.

• Live IoT Feeds: Link telemetry APIs to equipment nodes, enabling the brain to trigger causal failure alerts based on active heat, pressure, or vibration fluctuations.

• Voice-Activated AR Client: Build an AR-overlay app for field technicians, projecting lock-out procedures directly onto equipment casings on the plant floor.